

Track-System – Glossary

Last updated: 23 Sep 2026 | The words that keep coming back in this project, each one in plain language

1. The big picture – who talks to whom

FRONTEND (in the browser) **BACKEND** (on the computer) **DATABASE**

React + Vite ASP.NET Core (C#) SQL Server

| | |

|--- axios sends HTTP req ---> endpoint /api/... |

| |--- EF Core: C# turns to SQL --->|

|<----- JSON answer -----|<----- rows -----|

The frontend NEVER talks to the database directly. Always through the backend.

2. The platform words

Word	What it means	Where we meet it
.NET	Microsoft's platform for running C# programs, like Node.js is for JavaScript. It includes the compiler, the libraries and the runtime.	dotnet --version
ASP.NET Core	The part of .NET for building web servers and APIs.	Our whole backend
SDK / Runtime	SDK = the tools to <i>build</i> code. Runtime = only what is needed to <i>run</i> it. A developer needs the SDK.	We installed the SDK
Kestrel	The actual web server inside ASP.NET Core. It listens on the port and receives the requests. You never write it, it is just there.	The line "Now listening on: ..."
NuGet	The library store for .NET. Every PackageReference in the .csproj is downloaded from there.	dotnet restore
npm	The same idea for JavaScript. The frontend's package.json is its shopping list.	npm install
Vite	The tool that runs the React project while developing and builds it for production. It gives the frontend its own address, usually port 5173.	npm run dev

3. The web words

Word	What it means
HTTP request	A message the browser sends to a server. It carries a method (GET/POST/...), an address, optional headers and optional body data.
GET / POST / PUT / DELETE	GET = give me data. POST = here is new data (for example login details). PUT = update something existing. DELETE = remove it.
Endpoint	One address on the server plus the code that answers it. Example: /api/health.
API	The whole collection of endpoints. The menu of everything the backend can do for the frontend.
REST	A common style for designing an API: the addresses are nouns (/api/employees) and the methods are the verbs (GET, POST).
JSON	The text format both sides speak, for example {"status":"ok"}. C# objects become JSON on the way out, JSON becomes C# objects on the way in.

Word	What it means
Status code	A number describing the result: 200 OK, 201 created, 400 bad request, 401 not logged in, 403 not allowed, 404 address not found, 500 server crashed.
localhost	"This computer". Not reachable from outside, which is exactly what we want while developing.
Port	The door number on the machine. 5163 = our API, 5173 = React. Two programs cannot use the same port.
axios	A JavaScript library, already installed in the frontend, that sends the HTTP request from React and gives back the JSON answer.
CORS	A browser security rule: a page served from port 5173 may not call port 5163 unless the server says it is allowed. Adding that permission is a backend step.

4. The database words

Word	What it means
SQL Server	Microsoft's database program. SQL Express is the free local edition.
SSMS	SQL Server Management Studio, the app where you look at the tables by hand.
Connection string	One line of text saying which server, which database and how to log in. Ours lives in appsettings.json.
Windows Authentication	Logging into the DB with your Windows user, so no password is written anywhere.
ORM	Object Relational Mapper: software that maps database rows to objects in code, so you stop writing SQL strings by hand.
EF Core	Entity Framework Core, Microsoft's ORM. It translates the C# we write into SQL and turns the returned rows back into C# objects.
DbContext	The EF Core class that represents the database. Our ApplicationDbContext inherits from it, and every query goes through it.
DbSet	One line inside the DbContext representing one table. Step 3 adds DbSet<Employee> Employees.
Model / Entity	A plain C# class whose properties mirror the columns of one table.
Migration	EF Core creating or changing tables out of your C# classes. We do NOT use it: our tables already exist, so we go the other way around.
Provider	The package that knows the dialect of one specific database. Ours is Microsoft.EntityFrameworkCore.SqlServer.

5. The C# and code words

Word	What it means
Namespace	The address of a class so other files can find it with using Ours is TrackSystem.Api.Data.
Class / Property	A class is a blueprint. Its properties are the fields inside it, which will match the table columns.
Inheritance (:)	class ApplicationDbContext : DbContext means it receives all of DbContext's abilities for free.

Word	What it means
Lambda (=>)	A small function written inline and kept for later. The code inside MapGet is a lambda: it runs on each request, not at startup.
async / await	Talking to a DB takes time. await means "wait for the answer without freezing the server", so other requests keep being served.
Dependency Injection	You ask for something in the parentheses and the server hands you a ready one. That is why we never write <code>new ApplicationDbContext(...)</code> .
Service registration	Teaching the server how to build those objects. Anything registered with <code>builder.Services</code> must be written before <code>Build()</code> .
Middleware	Code every request passes through on its way in (the CORS check, the login check, logging) before it reaches the endpoint.
Minimal API	The style we use: endpoints written directly in Program.cs. The older style is Controllers, which are separate class files.

6. The security words (coming in the next steps)

Word	What it means
Hashing	Turning a password into scrambled text that cannot be turned back. We compare hashes, so the real password is never stored.
BCrypt	The hashing library we will use. It is deliberately slow, which makes guessing attacks expensive.
Salt	Random characters added before hashing, so two people with the same password still get different hashes.
Authentication	Who are you? This is the login itself.
Authorization	What are you allowed to do? These are the permissions, for example the UserPermissions table.
JWT	JSON Web Token: a signed string the server gives you after a successful login. The frontend sends it back on every later request as proof of identity.
Token expiry	A JWT dies after a set time, so a stolen one is not useful forever.

7. Tools and workflow

Word	What it means
git add / commit / push	add = choose the changes. commit = save a snapshot locally with a message. push = upload it to GitHub.
.gitignore	The list of files git must ignore: bin, obj, node_modules and anything secret.
bin / obj	Folders .NET generates while building. Never edit them, never commit them.
Build vs Run	Build = turn the C# into a program. Run = start it. <code>dotnet run</code> does both.
Hot reload	React updates in the browser the moment you save. The backend does not: after every change you stop it with Ctrl+C and run it again.